

CIS 671 Final Project Proposal

Project Title: Global Electric Vehicle (EV) Adoption & Market Trends Visualization

Team Members:

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Dataset Description:

Source: Kaggle

Dataset: Global EV Sales 2010-2024 Dataset

Attributes:

region: Indicates the name of the country or geographical region where EV data is collected. Example values include USA, Germany, China, etc.

category: Shows whether the data is historical or projected:

Historical: Refers to actual past recorded data.

Projected: Refers to forecasted or estimated future data (not present in this dataset).

parameter: Specifies the type of metric recorded. It includes:

- EV sales: The number of electric vehicles sold in a given year.
- EV stock: The total number of electric vehicles in use (cumulative over the years).
- EV sales share: The percentage of EV sales compared to total vehicle sales in a given year.
- EV stock share: The percentage of EV stock compared to total vehicle stock.

mode: Defines the vehicle category, e.g., "Cars" specifying the type of vehicles included.

powertrain: Classifies the type of EV drivetrain:

- BEV (Battery Electric Vehicle): Fully electric vehicles powered only by battery.
- PHEV (Plug-in Hybrid Electric Vehicle): Vehicles that combine a gasoline engine with an electric battery.
- EV: General electric vehicles category without specifying BEV or PHEV.

year: Records the specific year (from 2010 to 2024) when the EV sales or stock data was collected.

unit: Denotes the measurement unit of the value:

Could be number of vehicles (for EV sales or stock).

Could be percentage (for sales share or stock share).

value: The actual recorded numeric value for the corresponding metric (e.g., number of EVs sold, percentage share).

Supplementary Dataset (Optional):

What We Intend to Do: If time permits, we will optionally analyze the relationship between EV adoption trends and CO2 emissions using a standard emissions dataset.

Target User & User Persona:

Users:

- Policymakers tracking EV policy impact.
- Automotive industry analysts.
- Environmental researchers.
- Public consumers interested in EV trends.

Target User Tasks:

- ***Analyze Global EV Sales Trends Over Time:*** Visualizing overall global sales growth year by year.
- ***Analyze Regional EV Sales Trends:*** Comparing EV sales growth patterns across different regions.
- ***Compare Country-Wise EV Penetration Levels:*** Identifying countries with highest/lowest cumulative sales share and adoption levels.
- ***Analyze Powertrain Type Trends (BEV vs. PHEV):*** Breaking down sales trends by EV type to observe market preferences.
- ***Filter & Customize Analysis:*** Allowing users to filter by year range, region, and EV type dynamically.
- ***(Optional) Explore Relationship Between CO2 Emissions & EV Adoption:*** If time permits, correlating EV sales trends with CO2 emission reduction.

Planned Visualization Components:

We have come up with the following visualizations. However, we shall not show everything and do as much as we can based on the time constraints.

- **Line Chart** – Global EV Sales Growth Over Time
Showing how global EV sales have grown year by year.
- **Bar Chart** – Regional EV Sales Trends Comparison
Comparing EV sales growth patterns across different regions.
- **Stacked Bar Chart** – Powertrain Type Analysis (BEV vs. PHEV)
Breaking down EV sales by powertrain types (Battery Electric Vehicles vs Plug-in Hybrid Electric Vehicles).
- **Box Plot** – Distribution of EV Sales Across Different Regions
Showing variability and spread in EV sales among regions.
- **Choropleth Map** – Country-Wise EV Penetration Levels
Highlighting EV penetration percentages by country visually on the world map.
- **Pie Chart** – Powertrain Market Share for Specific Year & Region

Displaying the percentage share of BEV vs. PHEV in selected regions and years.

- **Histogram** – Distribution of EV Stock Share Across Countries
Showing how EV stock shares are distributed among all countries.
- **Circle Packing** – Regional Cumulative EV Sales Comparison
Visualizing cumulative EV sales across regions using nested circles sized by sales volume.
- **Parallel Coordinates Plot** – Multi-Attribute Regional Comparison
Comparing multiple metrics (sales, stock share, sales share) across regions.
- **Scatter Plot** – Correlation Between EV Sales & CO2 Emissions (Optional)
Visualizing the relationship between EV adoption and CO2 emissions reduction.
- **Dual-Axis Line Chart** – EV Adoption vs. CO2 Emissions Trends (Optional)
Showing EV sales growth alongside CO2 emissions trends over time.

Tools & Technologies:

- **Visualization Tool:** D3.js
- **Data Preprocessing:** Python (Pandas)
- **Data Sources:** Kaggle Global EV Sales Dataset, Optional CO2 dataset

Conclusion: This project will deliver an interactive dashboard visualizing EV adoption trends globally, offering insights into regional patterns, country-level penetration, powertrain preferences, and optional environmental impact analysis. The visualizations will feature multiple interactive components to support user-driven exploration and analysis, fulfilling the project proposal requirements.